Fast Fourier Transform (FFT)

Method, mathematics and a physical example

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October 25, 2026



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From the Fourier transform to the DFT

Continuous Fourier transform:

$$\hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-2\pi i f t}\, dt$$

A signal sampled at N points, spaced Δt apart ($f_s = 1/\Delta t$): $x_n = x(n\,\Delta t)$, $n = 0, \ldots, N-1$. Discrete Fourier transform (DFT):

$$X_k = \sum_{n=0}^{N-1} x_n\, e^{-2\pi i\, kn/N}, \qquad k = 0, \ldots, N-1$$

- Frequency of each bin: $f_k = k\, f_s/N$
- Resolution: $\Delta f = f_s/N = 1/T$ (total time of the record)
- Only $k \leq N/2$ is independent for real signals (Nyquist: $f_{\max} = f_s/2$)

The DFT in compact form

With $W_N = e^{-2\pi i/N}$:

$$X_k = \sum_{n=0}^{N-1} x_n \, W_N^{kn}, \qquad W_N = e^{-2\pi i/N}$$

- Matrix form: $\mathrm{X} = F\,\mathrm{x}$, with $F_{kn} = W_N^{kn}$ $(N \times N)$
- Inverse: $x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \, W_N^{-kn}$

Key properties of W_N

$$W_N^N = 1 \qquad \text{(periodicity)}$$

$$W_N^{k+N/2} = -W_N^k \qquad \text{(symmetry)}$$

$$W_N^2 = W_{N/2} \qquad \text{(halving)}$$

The problem: the cost of the DFT

- Each of the N outputs is a sum of N terms $\Rightarrow N^2$ complex multiplications.
- Example: 1 s of audio at 44 100 Hz $\Rightarrow N = 44100$ and about 2×10^9 operations.
- Many products repeat the same values of W_N^{kn} (periodicity and symmetry).

The FFT exploits exactly this redundancy.

Idea: divide and conquer

Take N even and split the sum into even and odd samples:

$$X_k = \sum_{m=0}^{N/2-1} x_{2m}\, W_N^{2mk} + \sum_{m=0}^{N/2-1} x_{2m+1}\, W_N^{(2m+1)k} = E_k + W_N^k\, O_k$$

Both E_k and O_k are DFTs of size $N/2$:

$$E_k = \sum_{m=0}^{N/2-1} x_{2m}\, W_{N/2}^{mk}, \qquad O_k = \sum_{m=0}^{N/2-1} x_{2m+1}\, W_{N/2}^{mk}$$

One DFT of size N becomes two DFTs of size $N/2$ plus N extra operations.

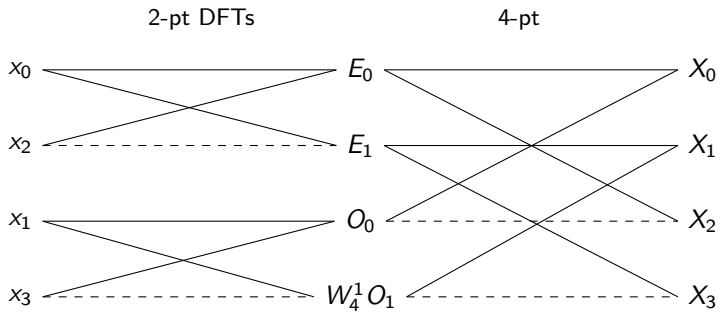
The butterfly

E_k and O_k have period $N/2$, and $W_N^{k+N/2} = -W_N^k$. Therefore, for $k = 0, \ldots, N/2 - 1$:

$$X_k = E_k + W_N^k \, O_k, \qquad X_{k+N/2} = E_k - W_N^k \, O_k$$

One product $W_N^k O_k$ gives **two** outputs: that is a *butterfly*.

Butterfly diagram for $N = 4$



Inputs in bit-reversed order; $W_4^0 = 1$, $W_4^1 = -i$; dashed lines multiply by -1.

Recursion and complexity

- Repeat the split until blocks have size 1 (the DFT of one point is the point itself).
- $\log_2 N$ stages, each with $N/2$ butterflies: $\dfrac{N}{2}\log_2 N$
- Equivalent recurrence: $T(N) = 2T(N/2) + \mathcal{O}(N) \Rightarrow \mathcal{O}(N \log N)$.

N	DFT (N^2)	FFT ($\frac{N}{2}\log_2 N$)	speed-up
$2^{10} = 1024$	$1\,048\,576$	$5\,120$	$\approx 205\times$
$2^{20} \approx 10^6$	1.1×10^{12}	5.2×10^6	$\approx 2 \times 10^5$

Worked example: $N = 4$

Samples of $x(t) = \sin(2\pi t/T)$ taken at $t = 0, T/4, T/2, 3T/4$: $\quad \mathrm{x} = (0,\ 1,\ 0,\ -1)$
Even samples (x_0, x_2) and odd samples (x_1, x_3):

$$E_0 = 0 + 0 = 0, \quad E_1 = 0 - 0 = 0, \qquad O_0 = 1 + (-1) = 0, \quad O_1 = 1 - (-1) = 2$$

Combine with $W_4^1 = -i$:

$$X_0 = 0, \quad X_1 = E_1 + W_4^1 O_1 = -2i, \quad X_2 = 0, \quad X_3 = E_1 - W_4^1 O_1 = 2i$$

- Only $k = 1$ and its mirror $k = N - 1 = 3$ are nonzero: one cycle in the window.
- Amplitude $A = 2|X_1|/N = 1$; the factor $-i$ means a *sine* (pure imaginary spectrum).

Physical example: two superposed vibrations

A sensor records the vibration of a structure with two modes:

$$x(t) = \sin(2\pi\, 50\, t) + 0.5\, \sin(2\pi\, 120\, t)$$

- $f_s = 1024$ Hz, $N = 1024$ samples ($T = 1$ s) $\Rightarrow \Delta f = 1$ Hz.
- Expected: peaks at $k = 50$ and $k = 120$, with amplitude $A_k = 2|X_k|/N$ equal to 1 and 0.5.
- The code uses `numpy.fft.fft`, an FFT implementation.

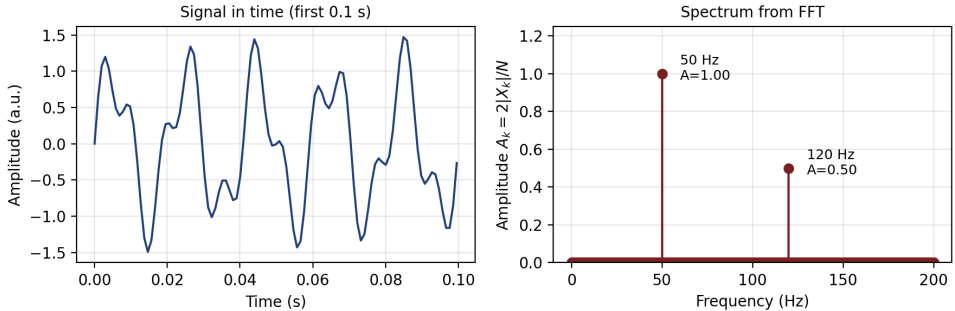
```python
import numpy as np

fs, N = 1024, 1024
t = np.arange(N) / fs
x = (np.sin(2*np.pi*50*t)
     + 0.5*np.sin(2*np.pi*120*t))

X = np.fft.fft(x)
f = np.fft.fftfreq(N, 1/fs)
A = 2*np.abs(X) / N      # amplitude

half = N // 2
print(f[:half][A[:half] > 0.1])
print(A[:half][A[:half] > 0.1])
```

Result



- Peaks at 50 Hz (A = 1.00) and 120 Hz (A = 0.50), as predicted. Other bins are below 10^{-14}.
- The FFT agrees with the direct N^2 DFT to 10^{-10} (rounding error).

Advantages and limitations

Advantages

- $\mathcal{O}(N \log N)$ instead of $\mathcal{O}(N^2)$
- Exact: same result as the DFT, not an approximation
- Fewer operations $\Rightarrow$ less rounding error
- Real-time analysis; optimized libraries (FFTW, numpy)
- Enables fast convolution, filtering, spectral PDE solvers

Limitations

- Assumes periodic signals $\Rightarrow$ spectral leakage (needs windows)
- No time localization (use STFT or wavelets)
- Uniform sampling; aliasing if $f_s < 2f_{\max}$
- Radix-2 needs $N = 2^m$ (padding, mixed radix, Bluestein)
- Resolution fixed by $\Delta f = 1/T$

Conclusions

- The FFT is an *exact and fast* way to compute the DFT.
- It splits a size-N DFT into even and odd parts recursively.
- Cost drops from N^2 to $\frac{N}{2}\log_2 N$.
- In the example it recovered frequencies and amplitudes of two vibration modes.

References

- J. W. Cooley and J. W. Tukey, *Math. Comput.* **19**, 297–301 (1965).
- M. Newman, *Computational Physics*, Ch. 7: Fourier transforms.
- W. H. Press et al., *Numerical Recipes*, Ch. 12–13.